

A Practical Guide to Named entity recognition

TOPIC -- NAMED ENTITY RECOGNITION

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Difficulties NER can have reference resolution ambiguities where the same name can refer to different entities of the same type. For example, "JFK" can refer to the former president or his son. The same name can refer to completely different types. "JFK" might refer to the airport in New York. "IRA" can refer to Individual Retirement Account or International Reading Association. This can be caused by metonymy. For example, "The White House" can refer to an organization instead of a location.

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Another field that has seen progress but remains challenging is the application of NER to Twitter and other microblogs, considered "noisy" due to non-standard orthography, shortness and informality of texts. NER challenges in English Tweets have been organized by research communities to compare performances of various approaches, such as bidirectional LSTMs, Learning-to-Search, or CRFs.

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Precision is the number of predicted entity name spans that line up exactly with spans in the gold standard evaluation data. I.e. when [Person Hans] [Person Blick] is predicted but [Person Hans Blick] was required, precision for the predicted name is zero. Precision is then averaged over all predicted entity names. Recall is similarly the number of names in the gold standard that appear at exactly the same location in the predictions. F1 score is the harmonic mean of these two. It follows from the above definition that any prediction that misses a single token, includes a spurious token, or has the wrong class, is a hard error and does not contribute positively to either precision or recall. Thus, this measure may be said to be pessimistic: it can be the case that many "errors" are close to correct, and might be adequate for a given purpose. For example, one system might always omit titles such as "Ms." or "Ph.D.", but be compared to a system or ground-truth data that expects titles to be included. In that case, every such name is treated as an error. Because of such issues, it is important actually to examine the kinds of errors, and decide how important they are given one's goals and requirements. Evaluation models based on a token-by-token matching have been proposed. Such models may be given partial credit for overlapping

matches (such as using the Intersection over Union criterion). They allow a finer grained evaluation and comparison of extraction systems.

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[Jim]Person bought 300 shares of [Acme Corp.]Organization in [2006]Time. In this example, a person name consisting of one token, a two-token company name and a temporal expression have been detected and classified.

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GATE supports NER across many languages and domains out of the box, usable via a graphical interface and a Java API. OpenNLP includes rule-based and statistical named-entity recognition. spaCy features fast statistical NER as well as an open-source named-entity visualizer. Hand-crafted grammar-based systems typically obtain better precision, but at the cost of lower recall and months of work by experienced computational linguists. Statistical NER systems typically require a large amount of manually annotated training data. Semisupervised approaches have been suggested to avoid part of the annotation effort. In the statistical learning era, NER was usually performed by learning a simple linear regression model on engineered features, then decoded by a bidirectional Viterbi algorithm. Some commonly used features include:

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Current challenges Despite high F1 numbers reported on the MUC-7 dataset, the problem of named-entity recognition is far from being solved. The main efforts are directed to reducing the annotations labor by employing semi-supervised learning, robust performance across domains and scaling up to fine-grained entity types. In recent years, many projects have turned to crowdsourcing, which is a promising solution to obtain high-quality aggregate human judgments for supervised and semi-supervised machine learning approaches to NER. Another challenging task is devising models to deal with linguistically complex contexts such as Twitter and search queries. There are some researchers who did some comparisons about the NER performances from different statistical models such as HMM (hidden Markov model), ME (maximum entropy), and CRF (conditional random fields), and feature sets. And some researchers recently proposed graph-based semi-supervised learning model for language specific NER tasks. A recently emerging task of identifying "important expressions" in text and cross-linking them to Wikipedia can be seen as an instance of extremely fine-grained named-entity recognition, where the types are the actual Wikipedia pages describing the (potentially ambiguous) concepts. Below is an example output of a Wikification system:

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with fewer tokens than desired (for example, missing the last token of "John Smith, M.D.") with more tokens than desired (for example, including the first word of "The University of MD") partitioning adjacent entities differently (for example, treating "Smith, Jones Robinson" as 2 vs. 3 entities) assigning it a completely wrong type (for example, calling a personal name an organization) assigning it a related but inexact type (for example, "substance" vs. "drug", or "school" vs. "organization") correctly identifying an entity, when what the user wanted was a smaller- or larger-scope entity (for example, identifying "James Madison" as a personal name, when it's part of "James Madison University"). Some NER systems impose the restriction that entities may never overlap or nest, which means that in some cases one must make arbitrary or task-specific choices. One overly simple method of measuring accuracy is merely to count what fraction of all tokens in the text were correctly or incorrectly identified as part of entity references (or as being entities of the correct type). This suffers from at least two problems: first, the vast majority of tokens in real-world text are not part of entity names, so the baseline accuracy (always predict "not an entity") is extravagantly high, typically >90%; and second, mispredicting the full span of an entity name is not properly penalized (finding only a person's first name when his last name follows might be scored as $\frac{1}{2}$ accuracy). In academic conferences such as CoNLL, a variant of the F1 score has been defined as follows:

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Lexical items: The token itself be labeled. Stemmed lexical items. Shape: The orthographic pattern of the target word. For example, all lowercase, all uppercase, initial uppercase, mixed case, uppercase followed by a period (often indicating a middle name), contains hyphen, etc. Affixes of the target word and surrounding words. Part of speech of the word. Whether the word appears in one or more named entity lists (gazetteers). Words and/or n-grams occurring in the surrounding context. A gazetteer is a list of names and their types, such as "General Electric". It can be used to augment any system for NER. They had been often used in the era of statistical machine learning. Many different classifier types have been used to perform machine-learned NER, with conditional random fields being a typical choice. Transformers features token classification using deep learning models.